A Time-of-Day-Aware Empirical Study of AWS Spot Instance Reliability for Short-Running Workloads

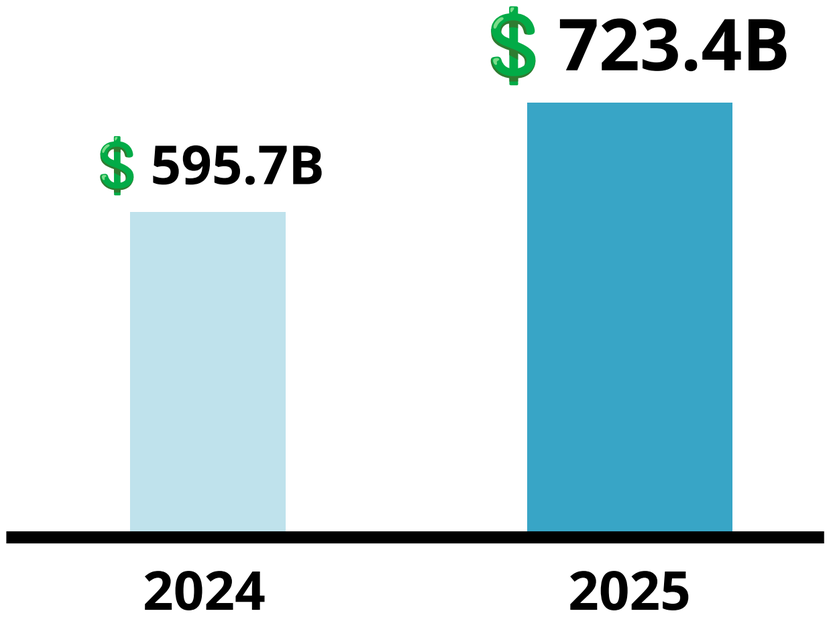
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Akihito Kohiga, Takahiro Koita

Graduate School of Doshisha University

1. Background

Growth of Cloud Usage and Its Cost Problem

Worldwide cloud spending is growing



https://www.gartner.com/en/newsroom/press-releases/2024-11-19-gartner-forecasts-worldwide-public-cloud-end-user-spending-to-total-723-billion-dollars-in-2025

The Problem of cost management

85%

Managing cloud spend is a problem

https://www.flexera.com/about-us/press-center/flexera-finds-cloud-value-is-rising-while-ai-waste-grows

Spot Instances

1. Background

Spot Instances

Use spare resources

at up to a 90% discount



Interrupted when spare resources run low



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resources run low

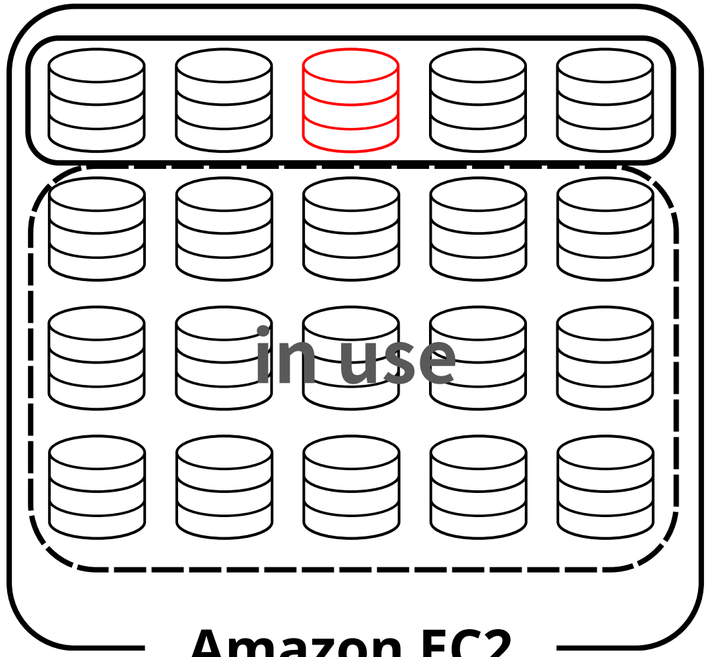


1. Background

Spot Instances

Use spare resources

at up to a 90% discount



Interrupted when spare

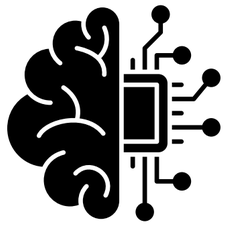
resources run low



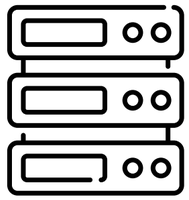
1. Background

Use Cases of Spot Instances

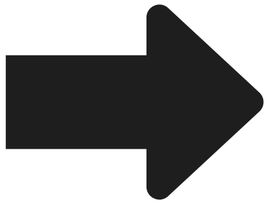
Conventional use



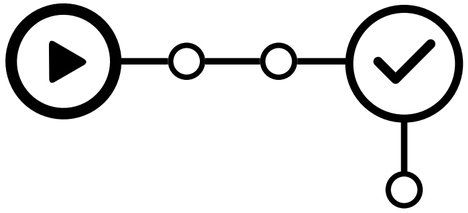
ML



Large-scale batch processing



New use



CI/CD

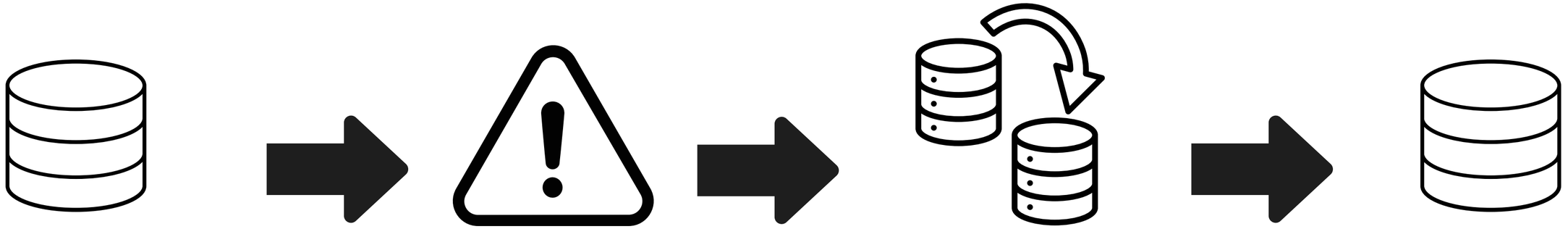


Automated testing

Short-running workloads (a few minutes to 1 hour)

1. Background

Limitations of the Existing Strategy



Spot Instances

Interruption notice

(2 minutes before)

Migration

(backup, etc.)

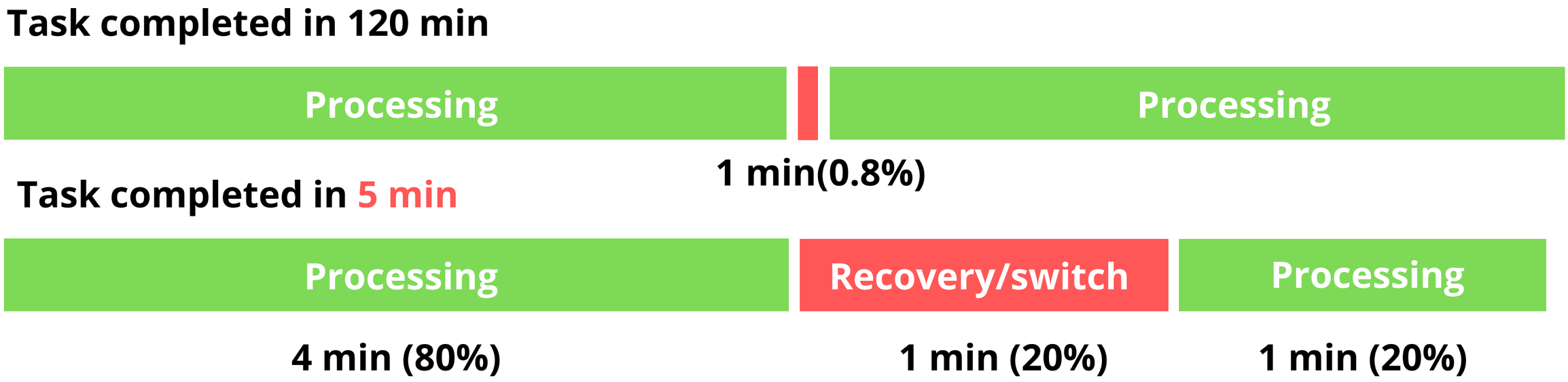
Another environment

A strategy that expects interruptions

1. Background

Short workloads require frequent recovery or switching,

which increases processing time and cost.



For jobs lasting a few minutes to an hour, checkpointing and restarting

take a large part of the total time.

1. 背景

For short-running workloads

We can complete rather than switch over

Task completed in 5 min

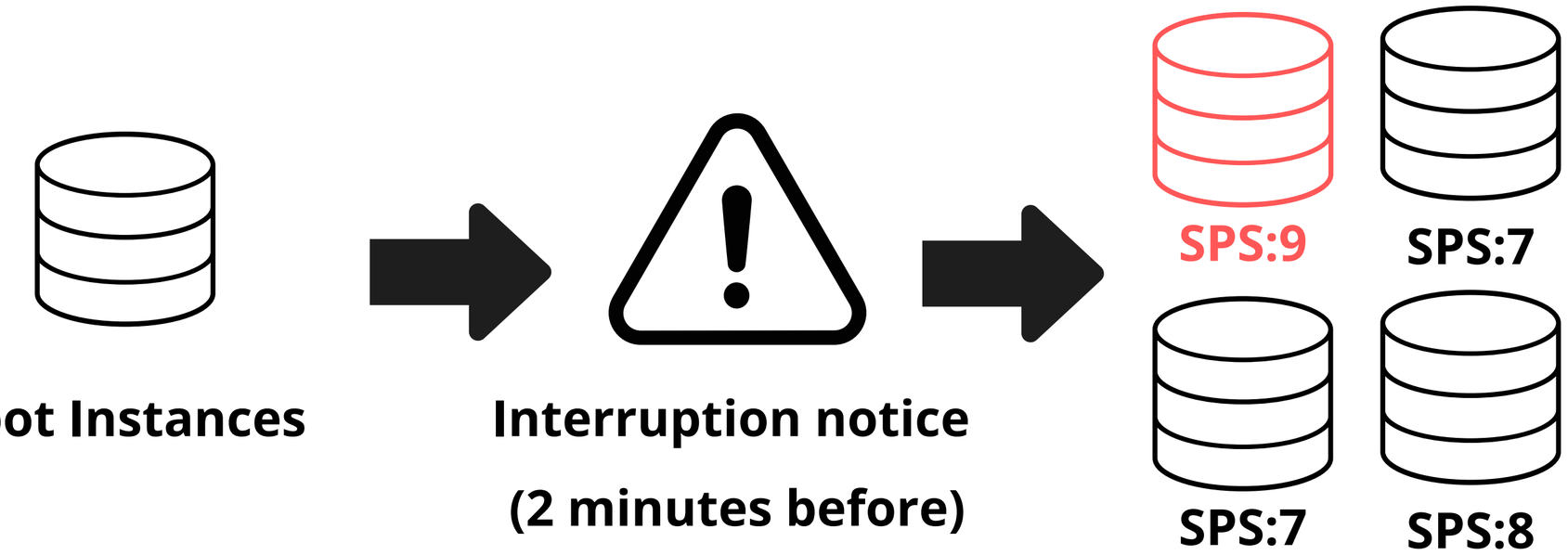


5 min (100%)

2. Related Work ①

Select a target instances using AWS SPS

※AWS SPS：A score from 1 to 10 showing the chance of launching a Spot Instances

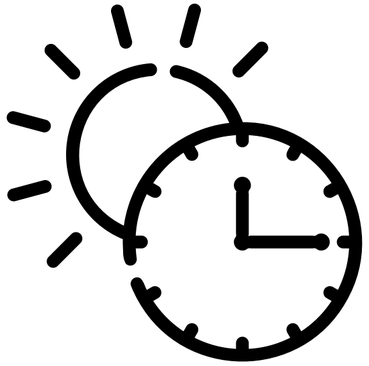


SPS only shows the chance of launch. It does not show whether the instances can keep running until the job finishes.

Katayama, D., Kasai, K., Koita, T.: Migration Destination Selection Algorithm for Spot Instances using SPS. In: 2022 IEEE International Conference on Big Data (Big Data), pp. 6690–6692. IEEE, Osaka (2022).https://doi.org/10.1109/BigData55660.2022.10020520

2. Related Work ②

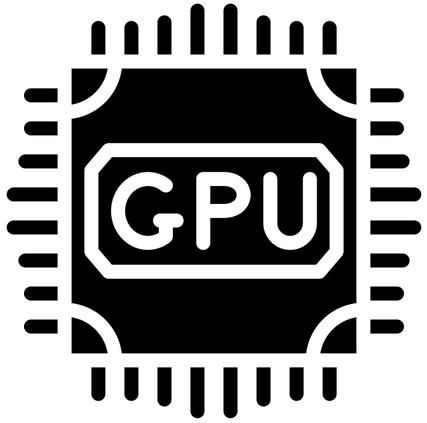
Spot Instances interruptions depend on the time of day, region, and instances type.



Higher during the day



Different across regions



Higher for GPU instances

Spot Instances interruption rates are not constant.

Lee, S., Hwang, J., Lee, K.: SpotLake: Diverse Spot Instance Dataset Archive Service. In: 2022 IEEE International Symposium on Workload Characterization (IISWC), pp. 242–255. IEEE (2022). https://doi.org/10.1109/IISWC55918.2022.00029

2. Limitations of Related Work

	Related Work①	Related Work②	This Study
Launch availability	○	×	×
Interruption risk	×	○	×
Task completion	×	×	○

Existing studies cannot determine whether a task will finish based on the Time zone, Region, and Instance type

3. Objective

Propose a strategy to choose

the best time zone, region, and instance type

for short workloads to finish successfully

4. Approach

step:1 Define a metric that quantifies completion feasibility

step:2 Measure on AWS across
Time zone × Region × Instance type × Deadline

step:3 Propose a selection strategy based on measured results

4. Approach

step:1 **Define a metric that quantifies completion feasibility**

step:2 **Measure on AWS across**
Time zone × Region × Instance type × Deadline

step:3 **Propose a selection strategy based on measured results**

4. Approach

step1: Define a metric that quantifies completion feasibility

$F_{comp}(d)$: The probability of surviving d minutes after launch

d	Assumed workload
5 min	Unit / Lint test
10 min	Build
30 min	ECE test・ETL
60 min	Short batch jobs・ML

4. Approach

step:1 Define a metric that quantifies completion feasibility

step:2 **Measure on AWS across**

Time zone × Region × Instance type × Deadline

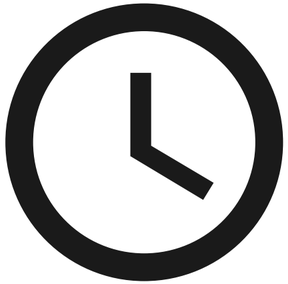
step:3 Propose a selection strategy based on measured results

4. Approach

step2:

Measure on AWS across

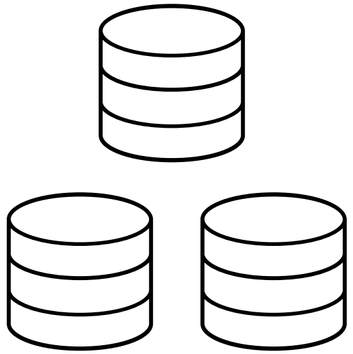
Time zone × Region × Instance type × Deadline



Time zone



Region



Instance type



Deadline(d)

4. Approach

step:1 Define a metric that quantifies completion feasibility

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Time zone × Region × Instance type × Deadline

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4. Approach

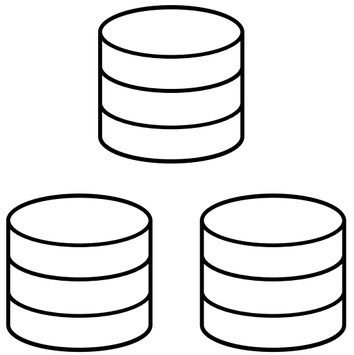
step3: Propose a selection strategy based on measured results



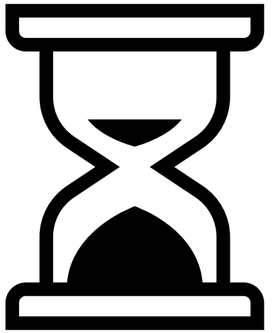
Time zone



Region



Instance type



Deadline(d)

Propose a strategy to choose
the best Time zone, Region, and Instance type
for short workloads to finish successfully.

5. Definition of the Metric

Probability that a spot instance survives d minutes after launch

$$F_{comp}(d) = \frac{N_{succ} - N_{int}(d)}{N_{succ}}$$

N_{succ} = Number of successfully launched spot instances

$N_{int}(d)$ = Number of Spot Instances interrupted within d minutes out of the N_{suss}

d = Workload (5, 10, 30, and 60 min)

5. Definition of the Metric

Ex: Probability that a spot instance survives 30 min after launch

100



Spot Instances
launched successfully

85 (15 interrupted)



Spot instances
surviving after 30 min

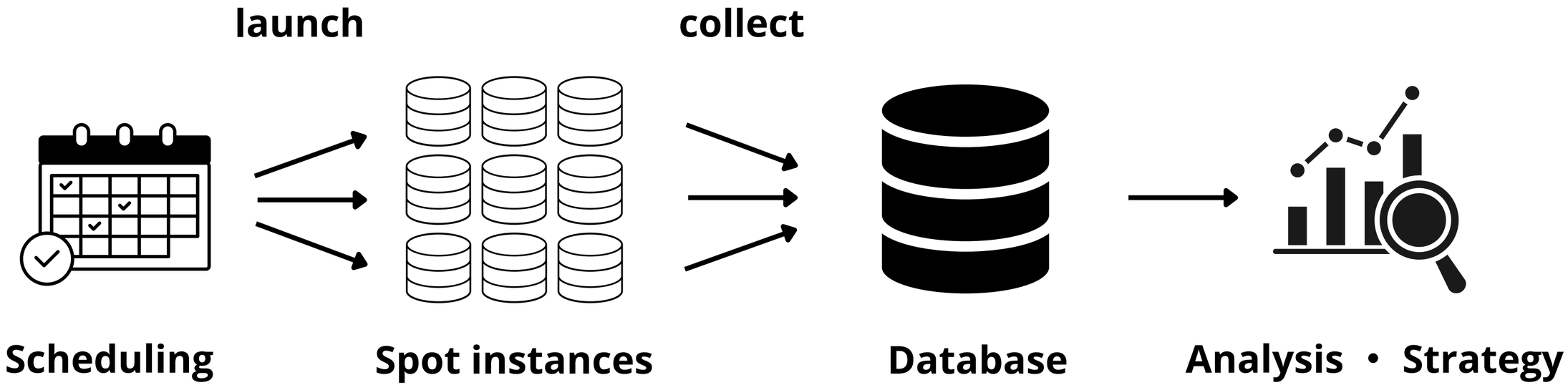
$$F_{comp}(30) = \frac{100 - 15}{100} = 0.85$$

85% of successfully launched Spot Instances

can run for 30 min and complete the workload.

6. Measurement Framework

Managing spot instances state in a databas



7. Experimental Setup

192 conditions（4 Time zone × 4 Region × 3 Instance type × 4 Deadline）



Time zone

Night (00 ~ 06)

Morning (06 ~ 12)

Afternoob (12 ~ 18)

Evening (18 ~ 24)



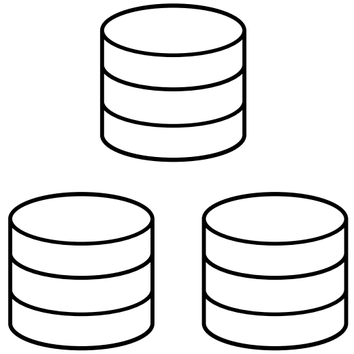
Region

N. Virginia

Ireland

Tokyo

Singapore

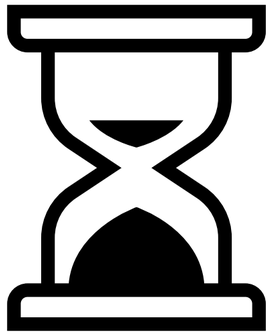


Instance type

c6a.large

g4dn.xlarge

t3.micro



Deadline(d)

5 min

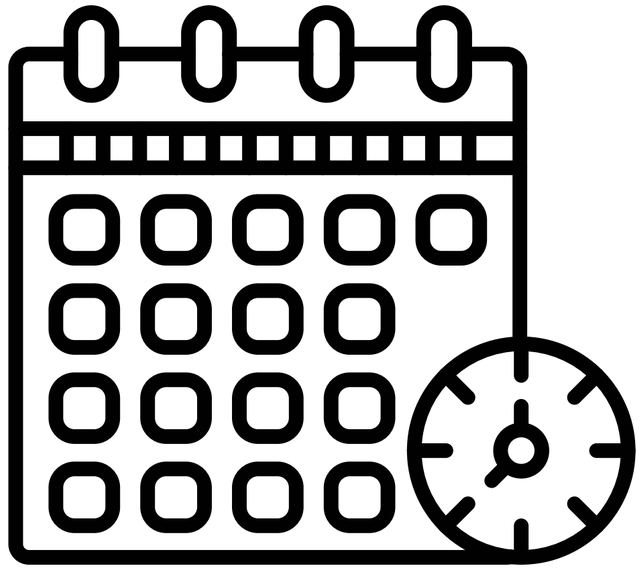
10 min

30 min

60 min

7. Experimental Setup

Observation period



05/19/2026 ~ 05/25/2025

(7 days)

8. Results and Strategy

$F_{comp}(60)$



8. Results and Strategy

Use Cases

Use Cases	d	Fixed	Selectable
CI/CD	5 min	-	All
Scheduled batch job (at the start of the workday)	10 min	Morning (Fixed Time zone)	Time zone Region
ETL (Run Locally)	30 min	Tokyo (Fixed Region)	Instance type Region
ML (GPU training)	60 min	g4dn.xlarge (Fixed Instance type)	Instance type Time zone

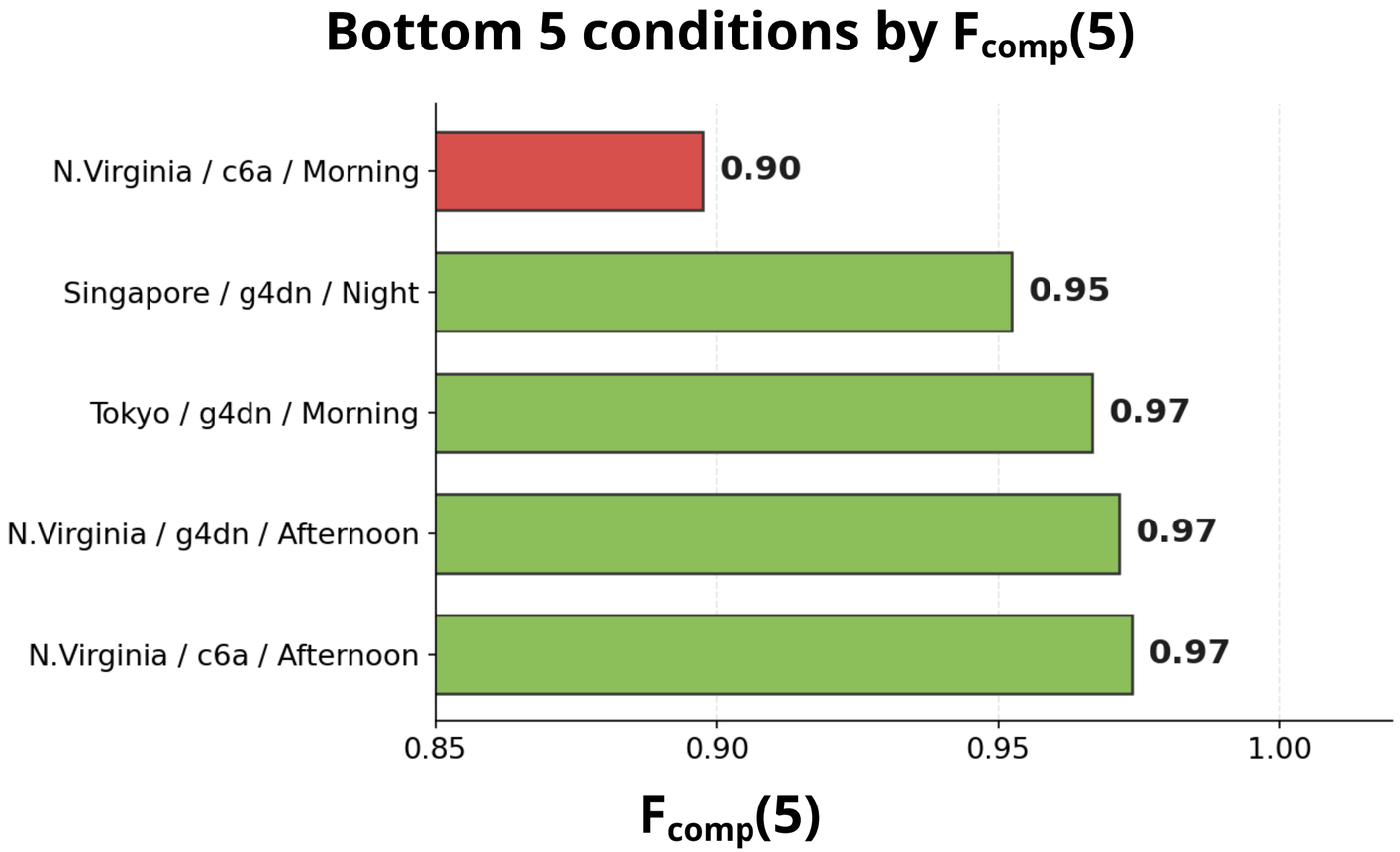
8. Results and Strategy

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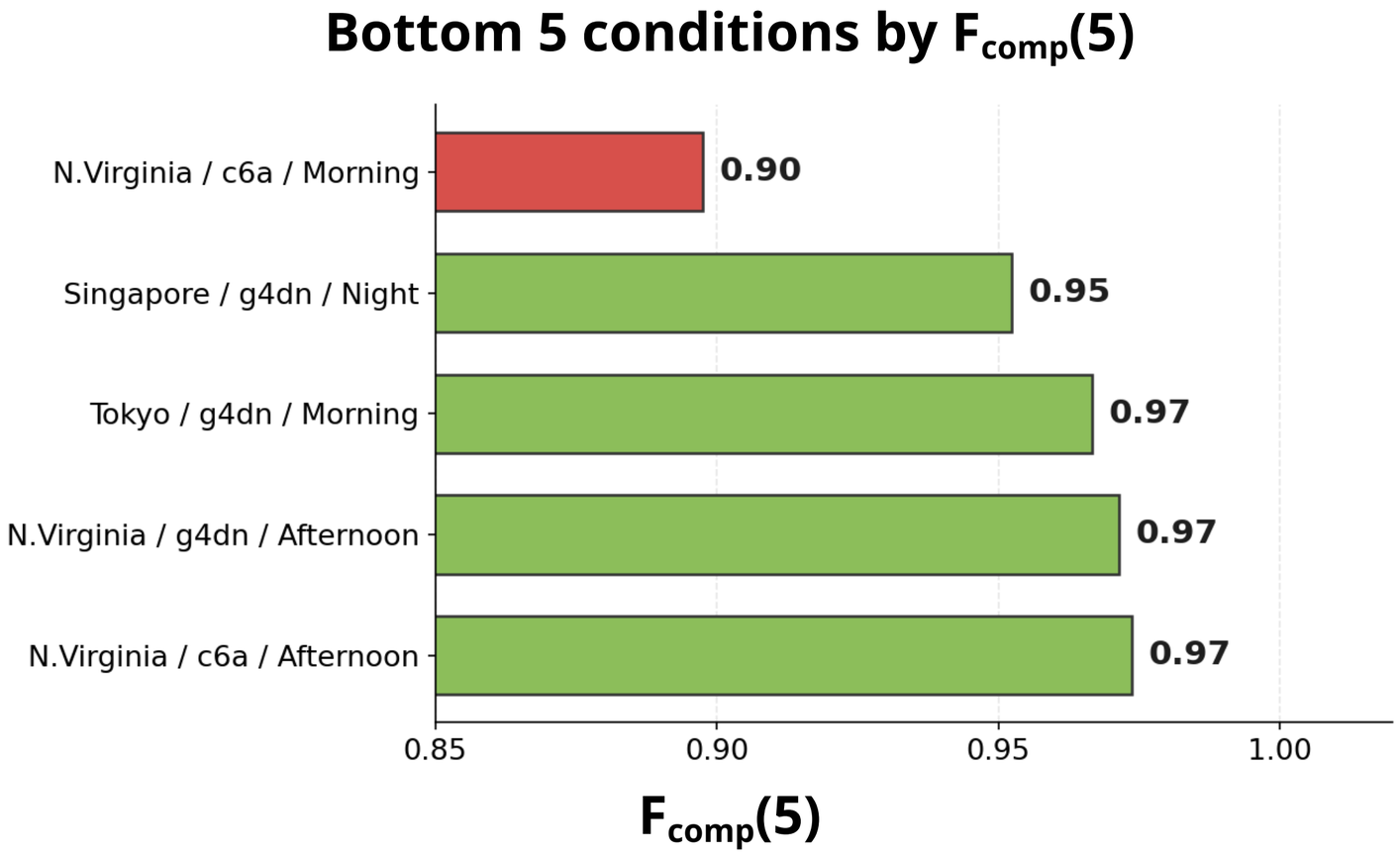
8. Results and Strategy

CI/CD



8. Results and Strategy

CI/CD



Strategy: Avoid c6a.large in N. Virginia during the Morning

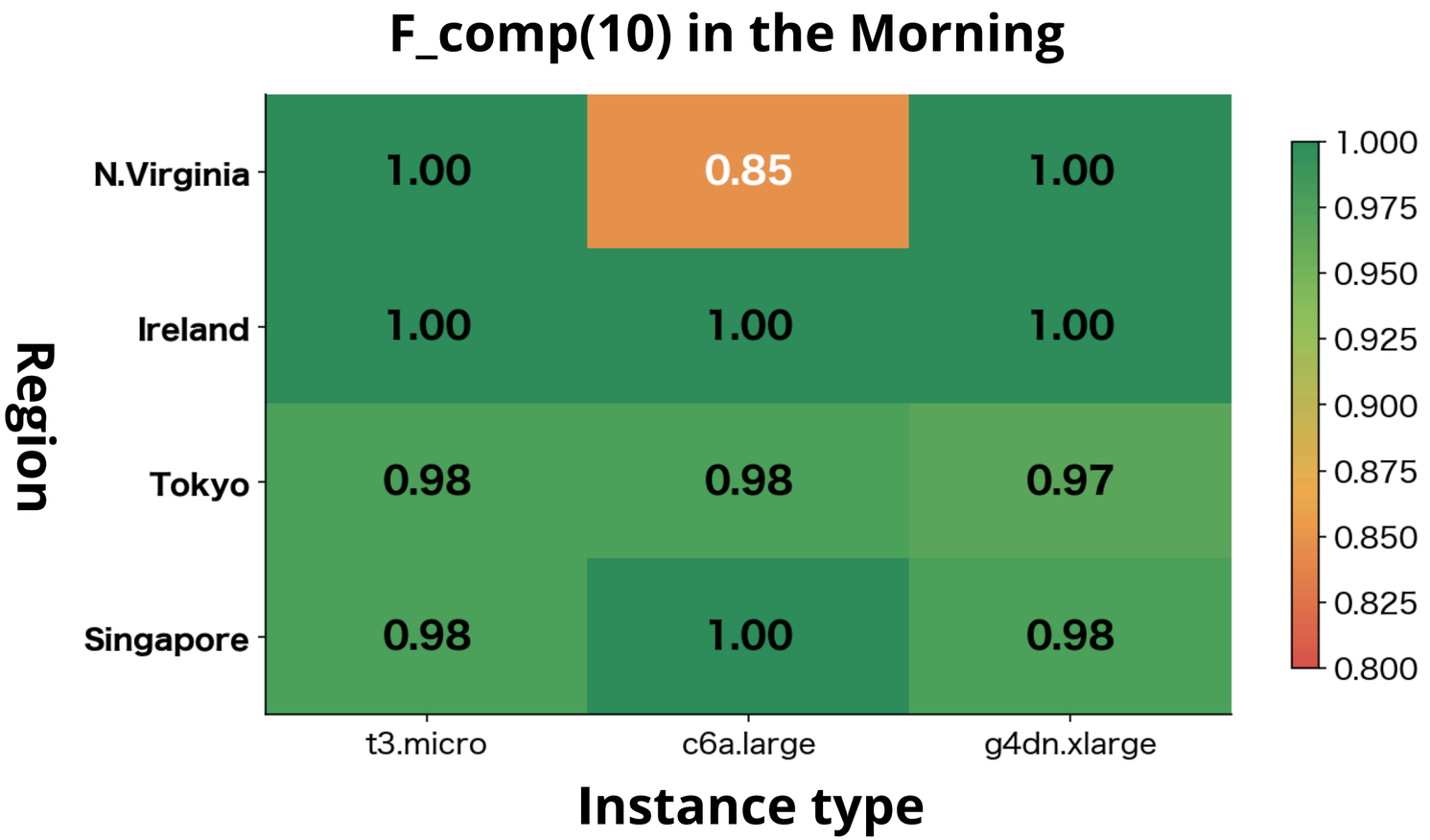
8. Results and Strategy

Use Cases

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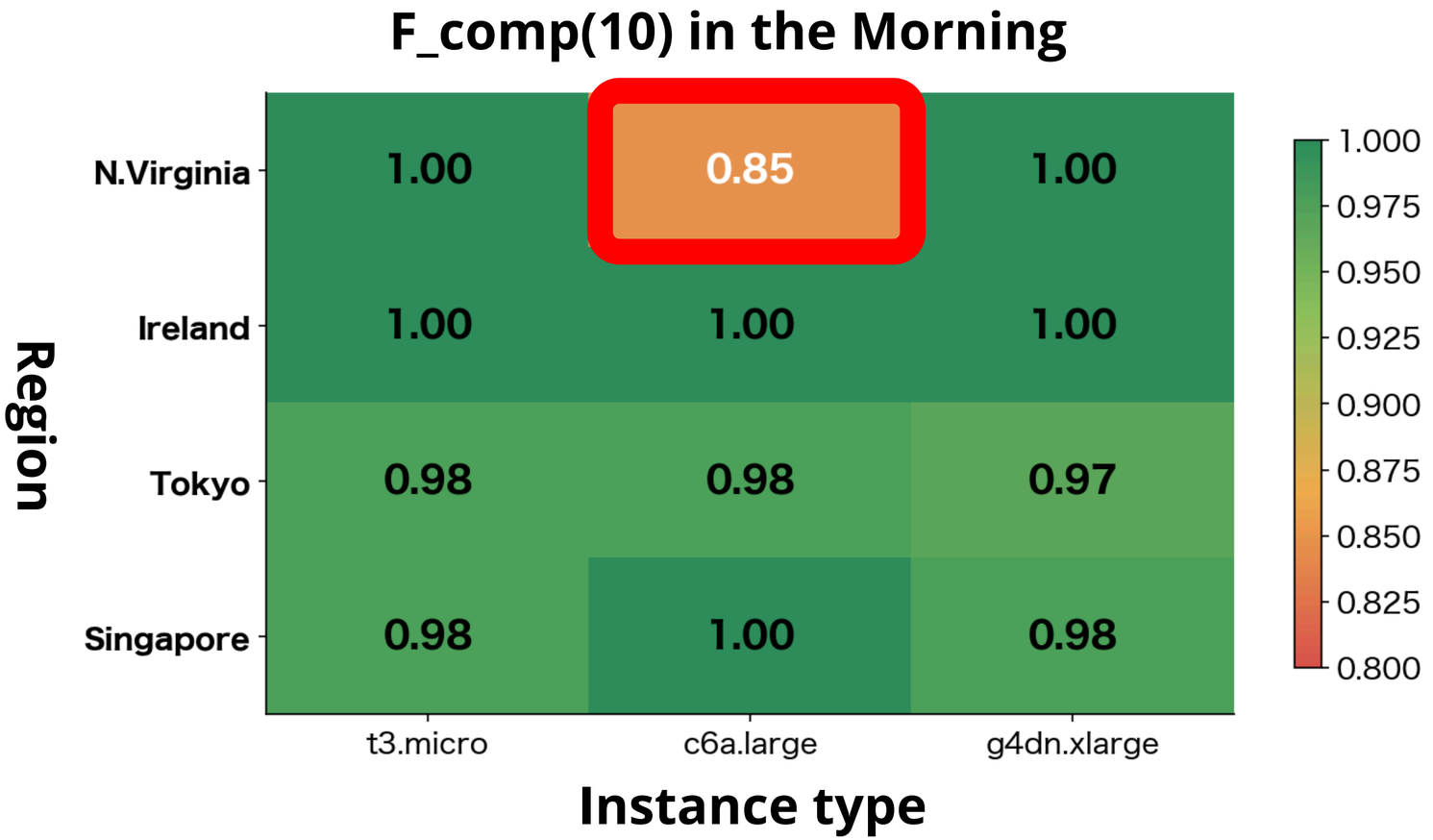
8. Results and Strategy

Scheduled batch job (at the start of the workday)



8. Results and Strategy

Scheduled batch job (at the start of the workday)



Strategy: Avoid c6a.large in N. Virginia

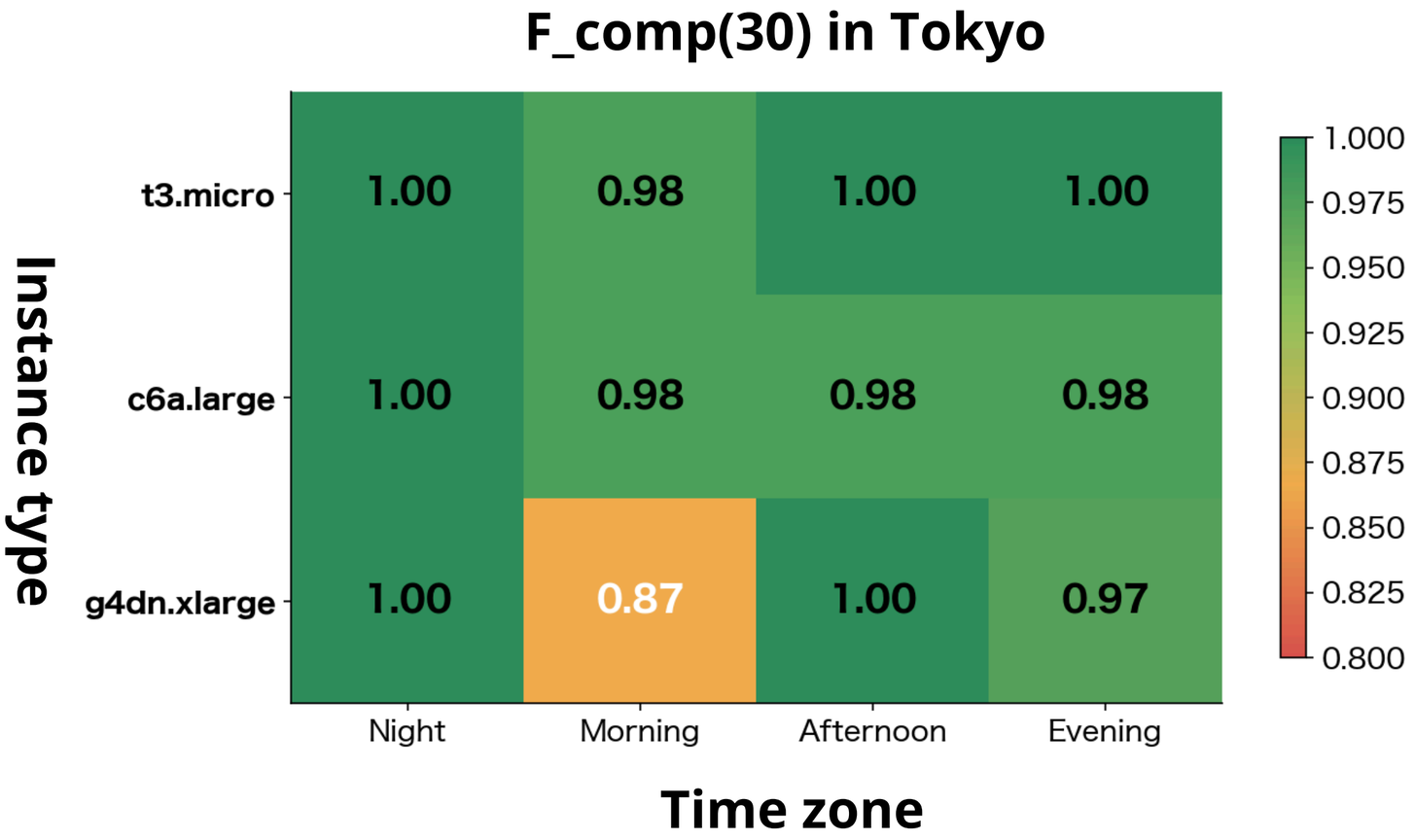
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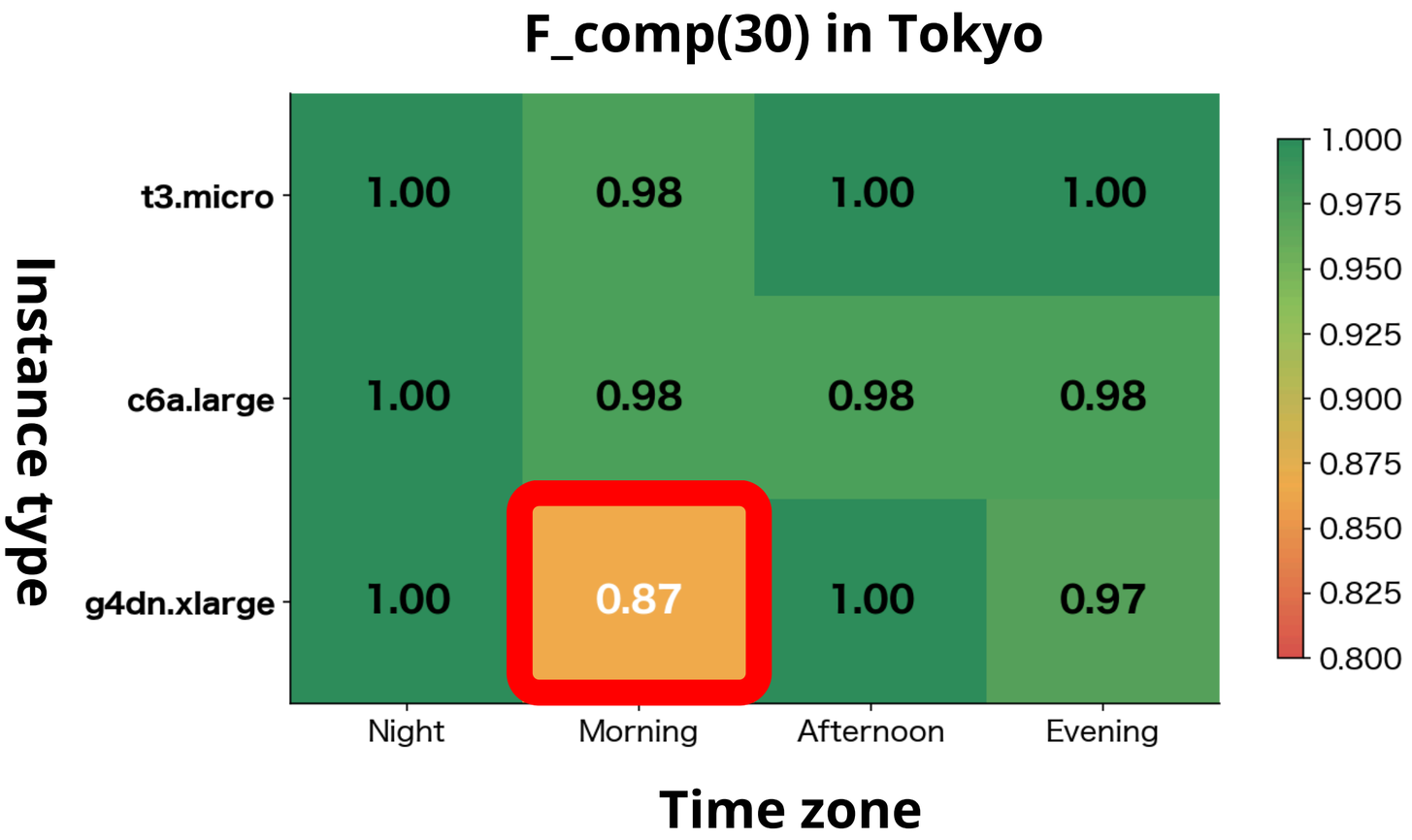
8. Results and Strategy

ETL(Run Locally)



8. Results and Strategy

ETL(Run Locally)



Strategy: Avoid g4dn.xlarge in the Morning

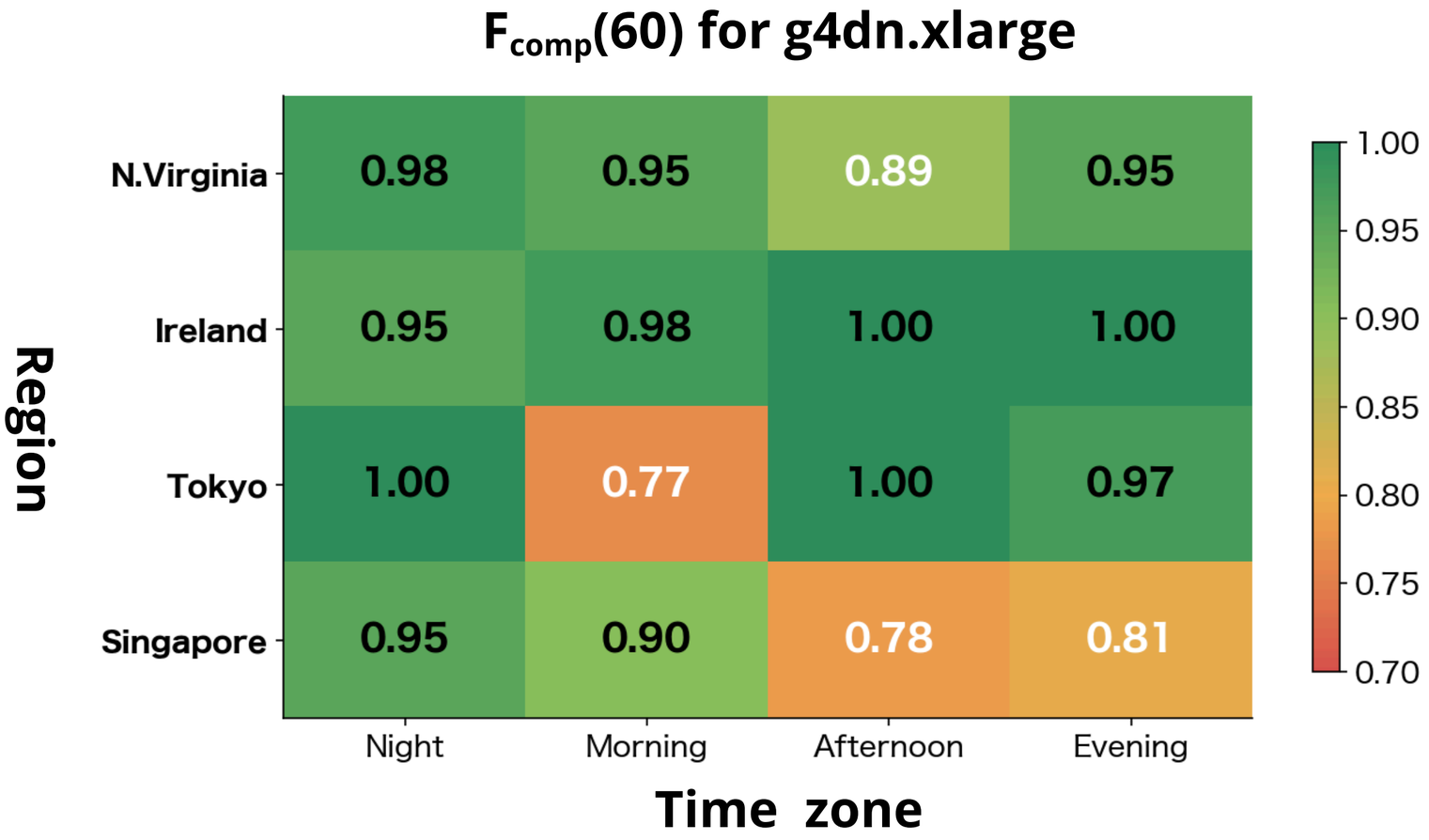
8. Results and Strategy

Use Cases

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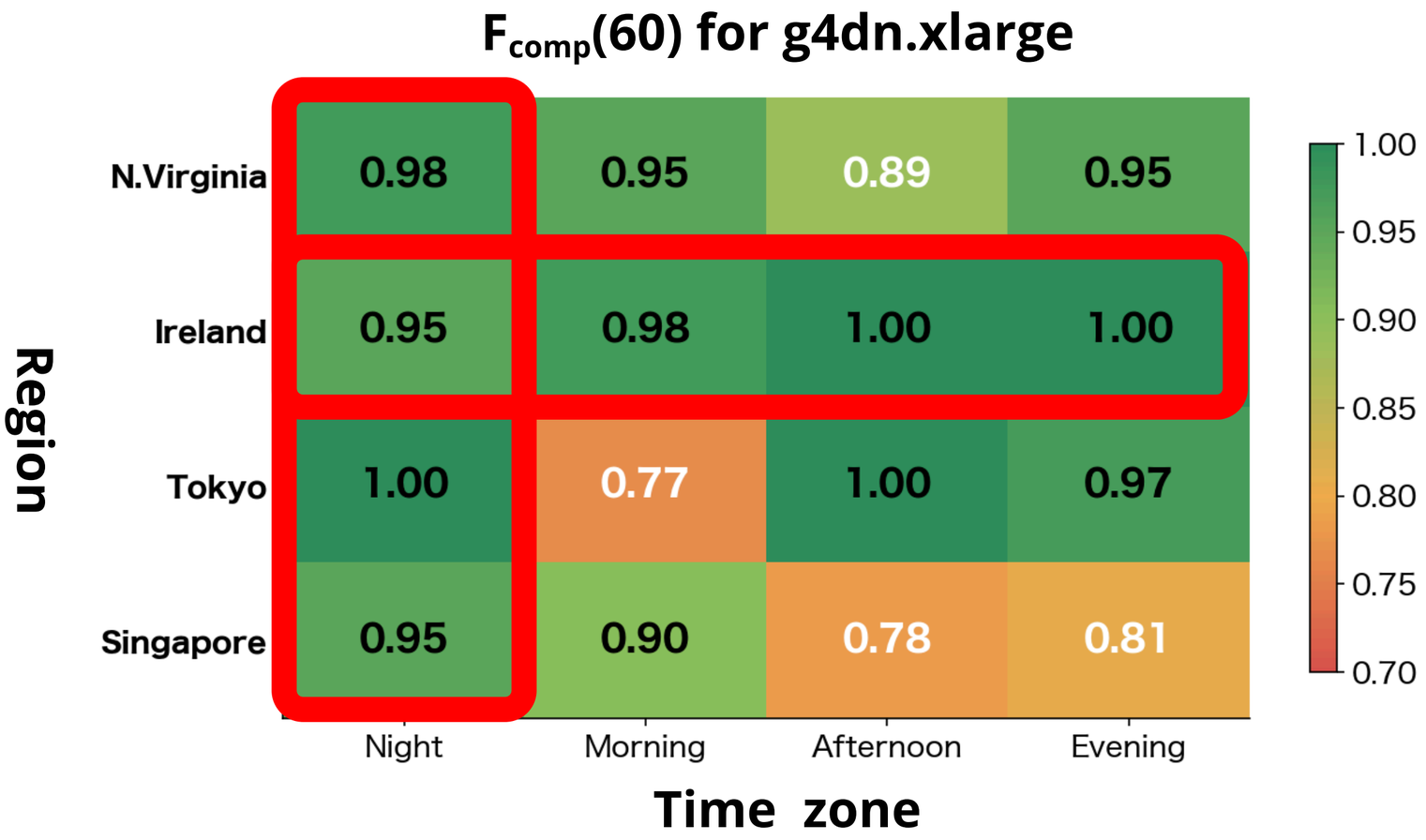
8. Results and Strategy

ML (GPU training)



8. Results and Strategy

ML (GPU training)



Strategy: Choose Night or Ireland

8. Results and Strategy

Use Cases

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ML (GPU training)	60 min	g4dn.xlarge (Fixed Instance type)	Instance type Time zone

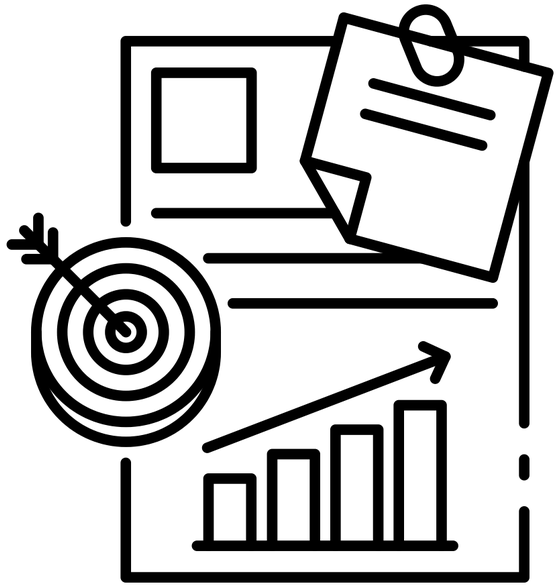
8. Results and Strategy

Use Cases

Use Cases	d	Strategy
CI/CD	5 min	Avoid c6a.large in N. Virginia during the Morning
Scheduled batch job (at the start of the workday)	10 min	Avoid c6a.large in N. Virginia
ETL (Run Locally)	30 min	Avoid g4dn.xlarge in the Morning
ML (GPU training)	60 min	Choose Night or Ireland

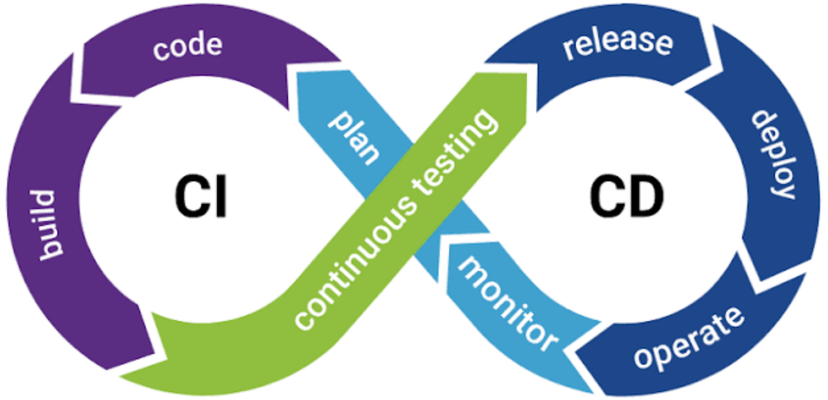
8. Future perspectives

Real-World Workload Evaluation



Strategy

(This Study)



Real-World Workload Evaluation

https://www.blackduck.com/ja-jp/glossary/what-is-cicd.html

Thank you